

axis, similar to an inkjet printer's functioning. This basic style is typically known as '3D printing', with more specialised names for other versions of this technique. An advanced version of this technique is known as 'PolyJet photopolymer' 3D printing which employs the traditional inkjet method of applying ink but uses a photopolymer liquid that solidifies when struck by UV light. The use of photopolymer allows for a variety of materials to be used in an assortment of colours and very high-resolution prints.

2. Stereolithography (SLA)

The next method is known as stereolithography. It is chemical based and relies on the combination of light sensitive chemicals and lasers. These chemicals are oriented in such a manner that when they're exposed to UV laser light they turn from liquid to solid. The 3D printers that use this method are designed to maneuver the UV laser across a thin surface of the chemical liquid in the design of the required object. As each layer solidifies, it is lowered and thinly submerged in the chemical liquid. The UV laser then creates another solid layer by moving across the liquid and so on until the final product is solidified and complete.

The stereolithography method has proven to provide a very high level of detailing and finishing on the surface of the objects created. The entire printing process takes places in the chemical liquid and the finishing involves separating the solidified 3D object from the pool of chemicals in a single flow. This was the first ever method of 3D printing that was invented in 1983 by Charles Hull.

3. Fused Deposition Modeling (FDM)

The third technique, known as 'Fused Deposition Modelling', is based on the use of molten material that becomes solid as it's layered on to the print surface. As the molten material is injected from the printer head, it creates the successive layers of the 3D design. This process continues until the product is fully created. This technique also uses food-based molten material such as cheese and chocolate to create complex-shaped food items. This is one of the most affordable types of 3D printers available in the market.

These types of printers use ABS and PLA plastics as well as biodegradable polymers which are organic in nature. The plastic additives used can also be dispensed as filaments from spools in a slightly augmented type of